JAVA MINI PROJECT | PROJECT PRESENTATION

TRAINING PROGRAM REGISTRATION & PARTICIPANT TRACKING SYSTEM

A simple, structured solution for managing registrations and attendance.

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01 | Introduction

- A Java-based application designed to register participants for training programs.
- Stores participant ID, name, email address, and selected training program.
- Provides participant lookup and a simple attendance-tracking feature.
- Uses a menu-driven console interface suitable for a beginner-level mini project.

02 | Problem Statement

- Manual registration can result in incomplete, misplaced, or inconsistent records.
- Finding an individual participant in paper-based records can take unnecessary time.
- Attendance information may be difficult to maintain accurately.
- A digital system can organize these tasks in a consistent and accessible workflow.

03 | Project Objectives

- Register participants and automatically assign a unique participant ID.
- Display all participants registered during the current session.
- Search for a participant using their assigned ID.
- Mark attendance and display the total number of registered participants.
- Apply core Java concepts to solve a practical administrative problem.

04 | Technologies Used

- Java — core programming language used to build the application.
- ArrayList — stores participant objects while the program is running.
- Scanner — accepts input from the keyboard.
- Classes and objects — represent participant records using object-oriented programming.
- Methods, loops, and switch-case statements — organize the program logic.

05 | Main System Modules

- Registration Module — accepts name, email, and training program details.
- Participant List — displays all stored participant records.
- Search Module — locates a record using its unique participant ID.
- Attendance Module — marks a registered participant as Present.
- Summary Module — displays the total number of registrations.

06 | System Workflow

- Step 1 — Launch the program and display the main menu.
- Step 2 — Select registration, viewing, search, attendance, or participant count.
- Step 3 — Enter participant details or the relevant participant ID.
- Step 4 — Validate the input and perform the selected operation.
- Step 5 — Return to the menu; exit when the user selects option 6.

07 | Sample Program Output

```
===== TRAINING PROGRAM REGISTRATION =====
1. Register Participant
2. View All Participants
3. Search Participant by ID
4. Mark Attendance
5. Show Participant Count
6. Exit

Enter your choice: 1
Enter participant name: Arun
Enter email address: arun@gmail.com
Enter training program name: Java Programming
Registration successful! Participant ID: 1001
```

08 | Advantages & Limitations

- Advantages — simple to operate, automatically assigns IDs, and supports quick searches.
- Attendance status is easy to review alongside participant details.
- Limitation — information is stored in memory and is lost when the program closes.
- The current version does not include permanent database storage, user login, or attendance dates.

09 | Future Enhancements

- Integrate MySQL to save participant details permanently.
- Add options to edit or delete participant records.
- Record attendance by date and calculate attendance percentages.
- Introduce secure login for administrators and trainers.
- Develop a graphical interface or web-based version.

10 | Conclusion

- The system organizes the basic process of training registration and participant tracking.
- It demonstrates practical Java concepts, including classes, objects, collections, and methods.
- It provides a foundation for a larger application with database and reporting features.
- Thank you.

THANK YOU

Questions & Discussion

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